

Data Analytics & AI for Business Decision-Making

A 13-class immersive journey from raw data to AI-powered insights. Master SQL, Power BI, Python, and Generative AI.

📅 13 Live Classes + Capstone 💻 Online & Instructor-Led 👤 Basic Computer Literacy



SQL



Power BI



Python



AI & Copilot



HTML/CSS/JS



Git & GitHub

Your Learning Journey

Six modules designed to take you from fundamentals to AI-powered analytics

MODULE 01

2 Classes

Analytics Mindset & Data Intelligence

Class 1 · The Data Analyst's Toolkit

Analytics vs BI vs DS Modern Data Ecosystem KPIs & Metrics Data-Driven Decisions Analytics Lifecycle
Career Paths Industry Use Cases

💡 *Workshop: Business Problem → KPI → Insight Framework*

Class 2 · Data Discovery & Insight Generation

Types of Data & Sources Structured vs Unstructured Data Collection Data Quality Ethics & Privacy
Patterns & Anomalies Data Storytelling

🔗 *Hands-on: Transforming Raw Data into Business Insights*

MODULE 02

2 Classes

Database & SQL Analytics

Class 3 · Database Foundations

DB Concepts & Architecture Tables & Relationships Primary & Foreign Keys ER Diagrams OLTP vs OLAP
Star Schema SQL Environment

🔗 *Exercise: Design a Sales & Customer Database*

Class 4 · SQL for Business Analytics

SELECT, WHERE, ORDER BY Aggregate Functions GROUP BY & HAVING JOINS Subqueries & CTEs
CASE Statements Window Functions KPI Reporting

★ *Portfolio Project: Customer, Sales & Transaction Analytics*

Power BI for Business Intelligence

Class 5 · Data Preparation & Modeling

Power BI Architecture Connect Multiple Sources Power Query (M) Data Cleaning Data Modeling

Star Schema

 *Build an analytics-ready data model*

Class 6 · Dashboard Design & Visualization

Visualization Best Practices KPI Cards Filters & Slicers Drill-through Dashboard UX Bookmarks & Buttons

Performance Reporting

 *Project: Business Performance Dashboard*

Class 7 · DAX & Advanced Analytics

Measures vs Columns Core DAX Time Intelligence Rolling Periods Growth Analysis Dynamic Reporting

KPI Automation

 *Project: Sales & Revenue Analytics Dashboard*

Class 8 · Data Storytelling & Publishing

Executive Dashboards Insight Presentation Performance Analyzer Mobile Reports Power BI Service

Scheduled Refresh Row-Level Security Governance

 *Project: Executive BI Dashboard (published)*

Python & Web Technologies for Analytics

Class 9 · Python Toolkit for Analysts

Anaconda & Jupyter NumPy Pandas Read Excel/CSV/JSON Data Cleaning Git & Version Control

Streamlit Basics

 *Hands-on: Automate Data Processing & Business Reports*

Class 10 · Forecasting, Analytics & Web Apps

KPI & Trend Analysis Matplotlib & Seaborn ARIMA, SARIMA, Prophet Model Evaluation XGBoost

HTML/CSS/JS Streamlit Integration

 *Project: Business Forecasting & Interactive Analytics App*

MODULE 05

2 Classes

AI-Powered Analytics & Automation

Class 11 · Generative AI for Analysts

AI in Modern Analytics

Data Troubleshooting

Diff Trends & Patterns

Cleaning & Transform

Reconciliation Support

Data segmentation & Insights

Governance & Risk

 Hands-on: AI-Based Portfolio & Performance Analysis

Class 12 · Analytics Automation & AI Agents

Workflow Automation

Reporting Pipelines

Power BI + Python + AI

Databricks Basic

Serverless Data Visualization

Future of Analytics

 Hands-on: Building an AI-Enhanced Analytics Workflow

MODULE 06

Finale

Capstone Project & Analytics Symposium

Class 13 · Analytics Showcase Day

End-to-end project: SQL Exploration, Python Analysis, Power BI Dashboards, Streamlit App, AI Insights, Business Recommendations

Symposium

Live demos · Peer review · Portfolio storytelling · Strategic roadmap presentation

 Project Walkthrough

 Collaborative Discussion

 Strategic Roadmap